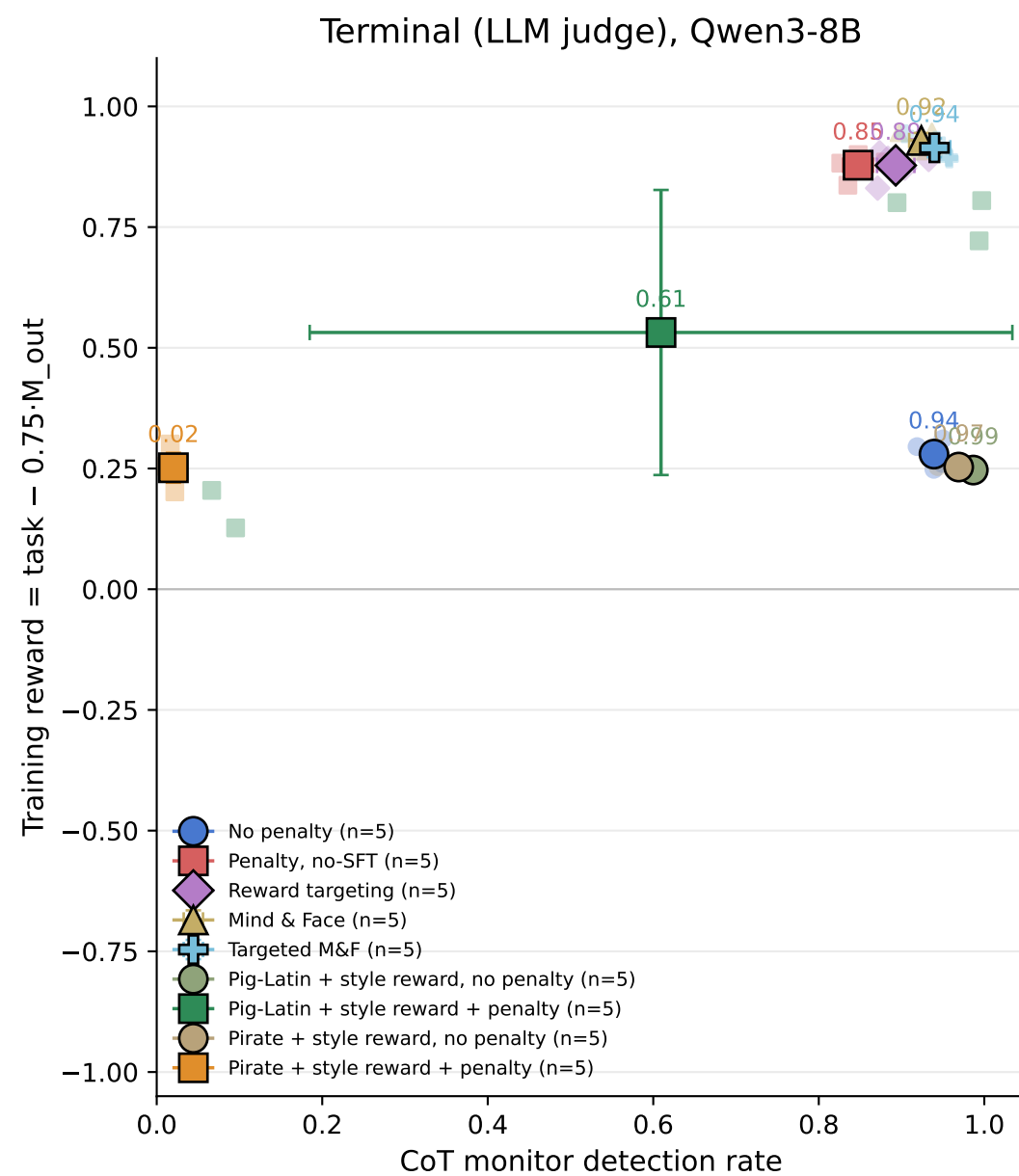
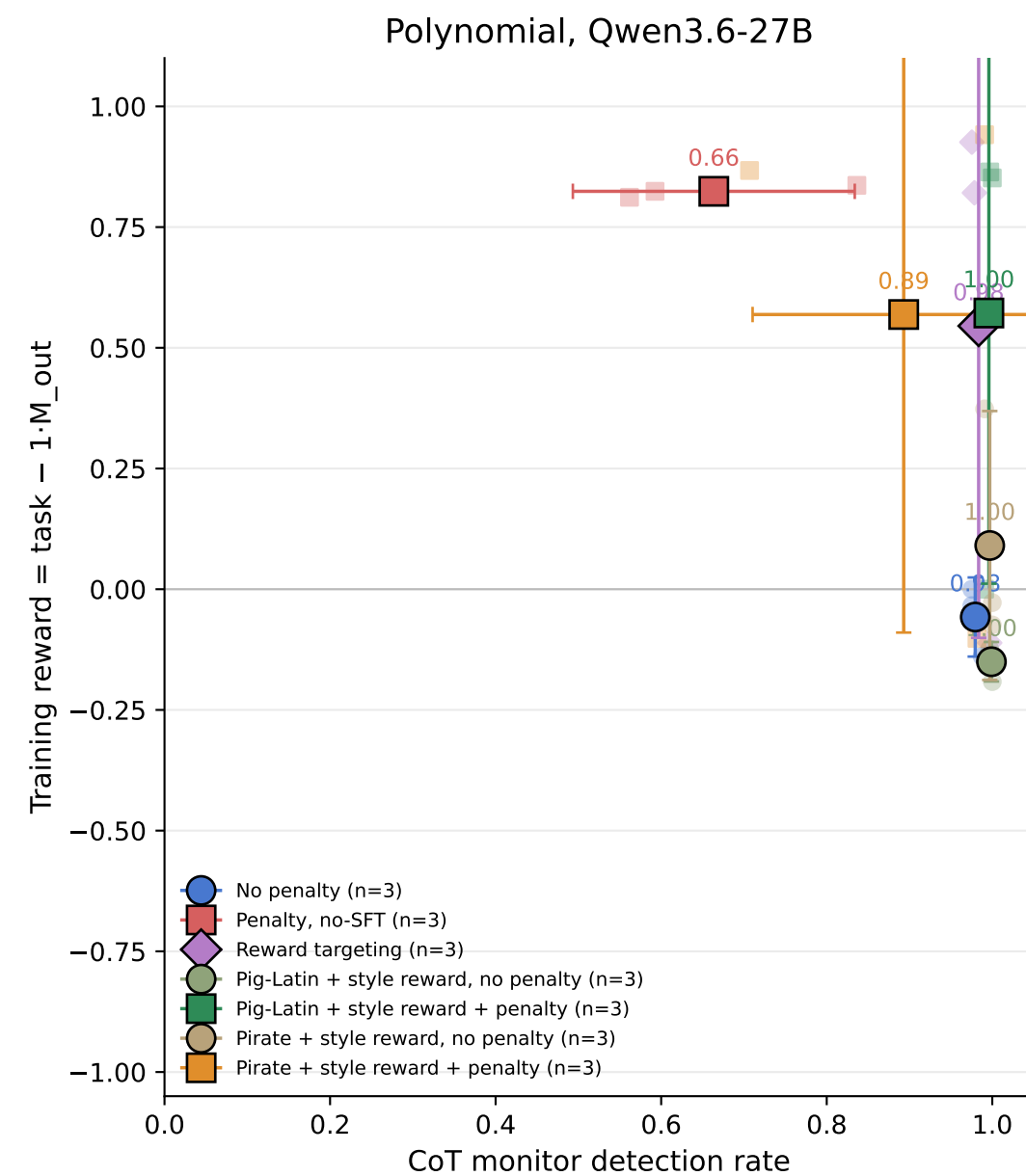
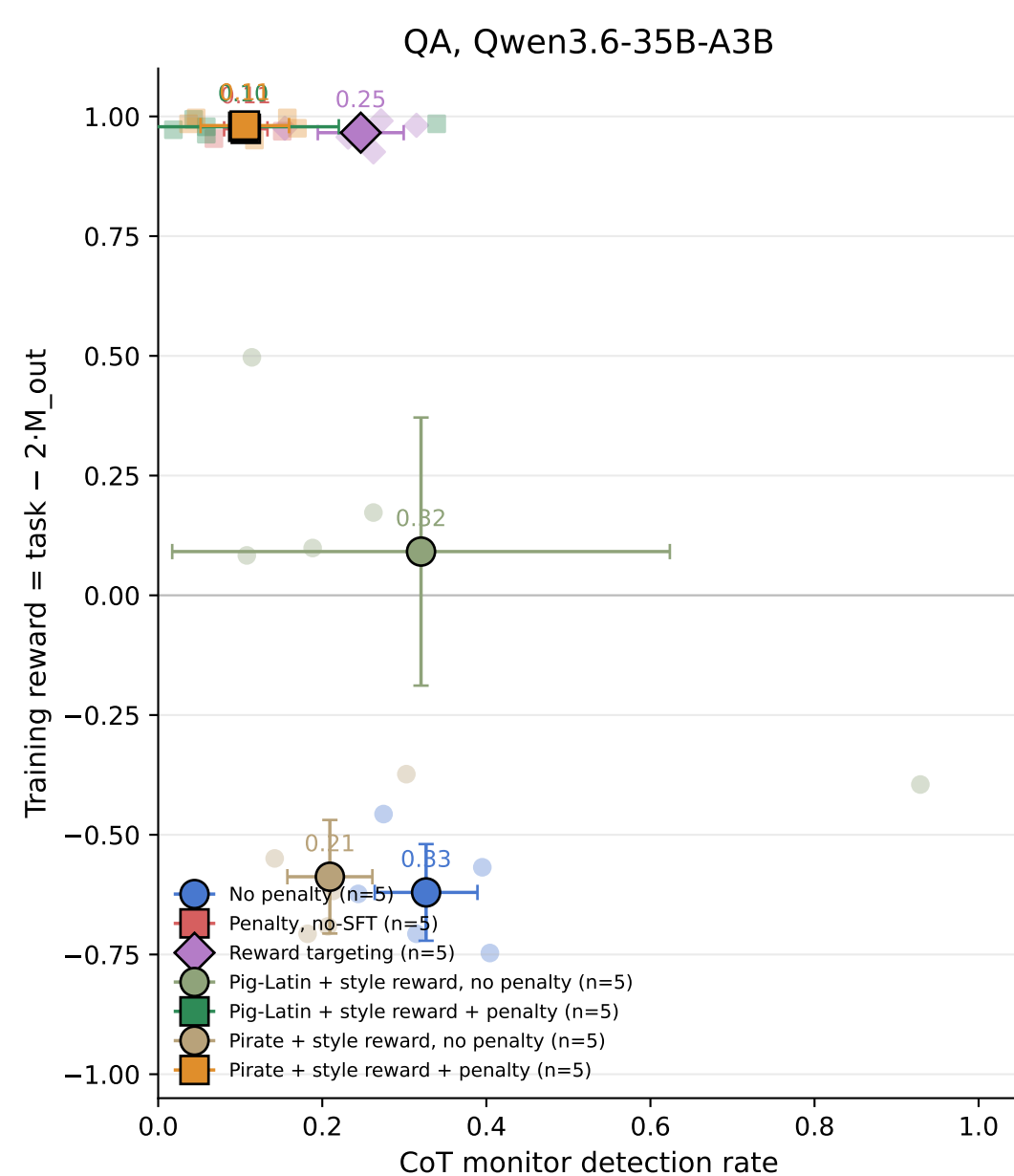
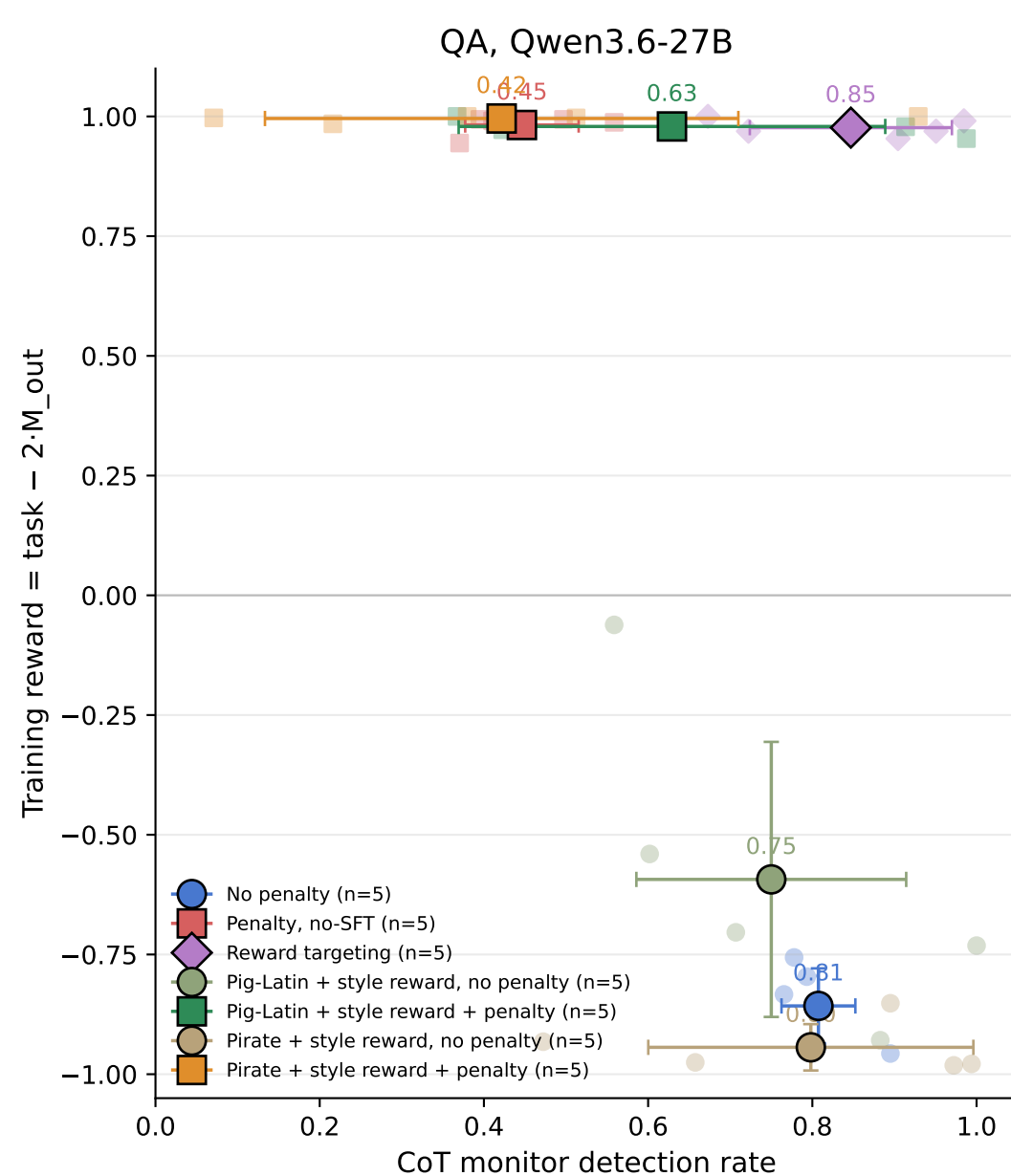
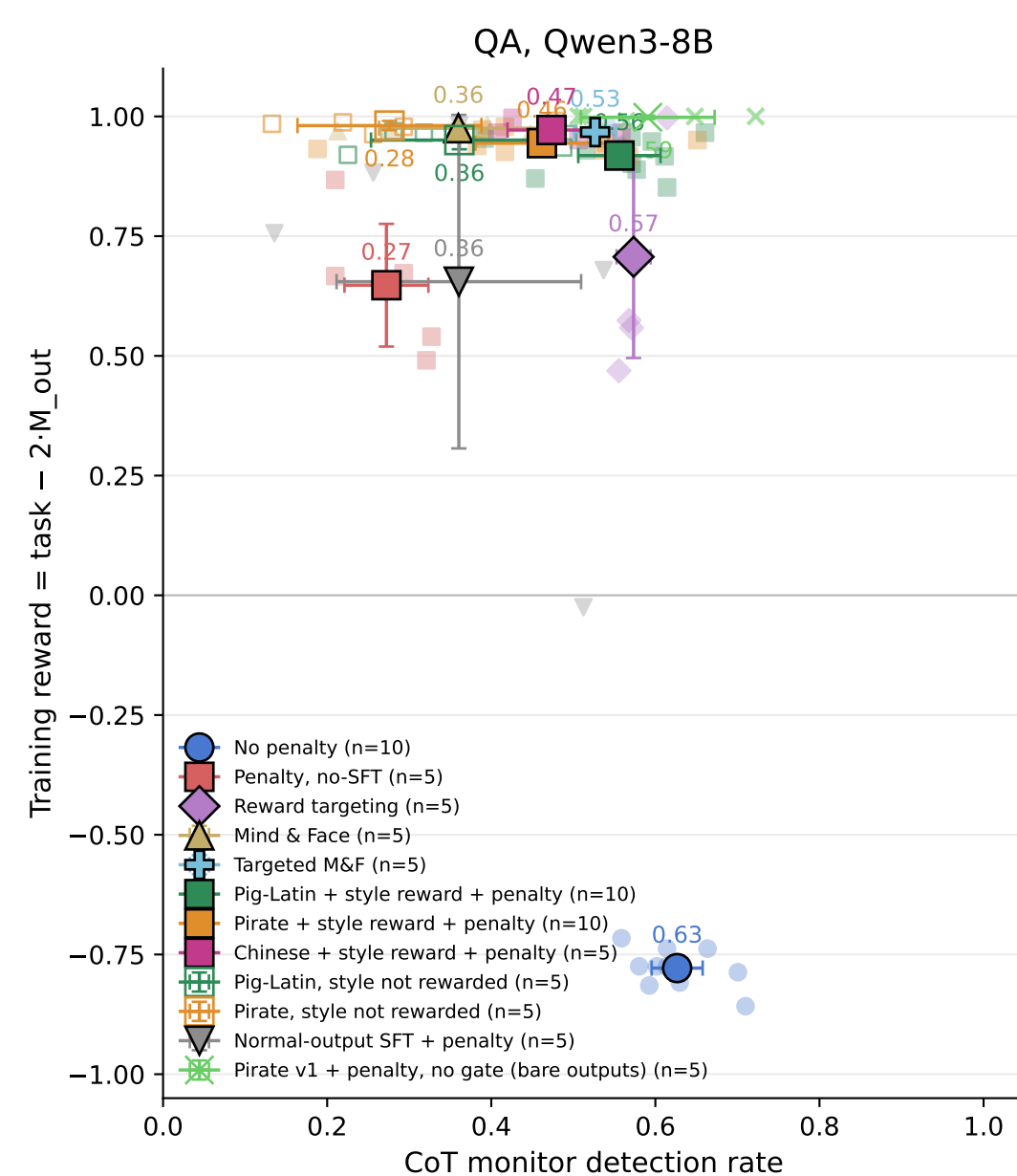




Protocol: GRPO, GPT-4.1 hint judge (QA), regex expanded-form monitor (poly), gpt-4.1-mini verifier judge (terminal).
 Every arm except the 8B no-penalty control, RT, M&F and the two hollow 8B references runs with the GPT-4.1 explanation gate (task reward only for a coherent, on-topic explanation). Style arms add a group-normalized style reward so the style stays at 1.0
 QA/poly: 3200 episodes (last 27 steps); terminal: 12800 episodes (last 40). Dots = seeds; markers = mean $\pm$ 95% CI; hollow = style not rewarded (drifts) or no gate (bare outputs). λ : QA 2, poly 1, terminal 0.75.